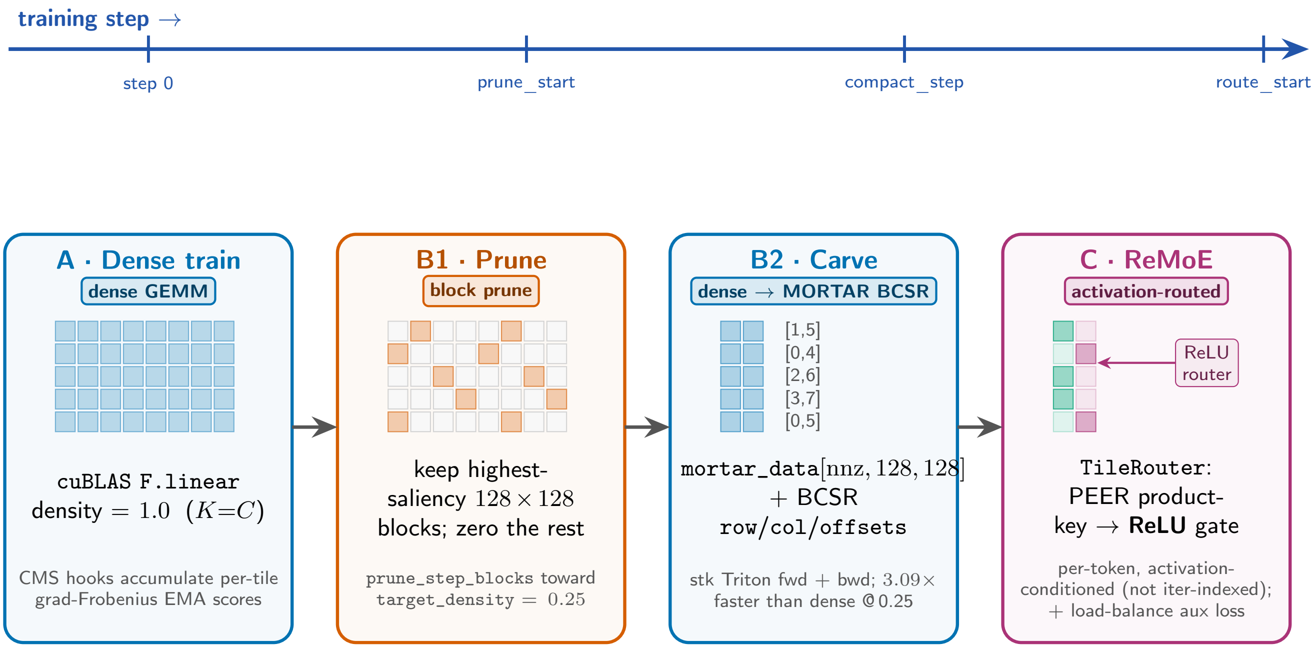
MORPH — Core-MLP Lifecycle: MORTAR · CMS Pruning · ReMoE



MORTAR BCSR: 128×128 blocks; $R_b{=}d_{\text{out}}/128$ block-rows, $C_b{=}d_{\text{in}}/128$ block-cols, nnz kept blocks, **density** $= \text{nnz}/(R_b C_b)$. **Scope:** all SwiGLU MLPs (`gate_up` + `down`) across **prelude** + **core** + **coda** (MORTAR-only — no scope knob); the core block is re-applied every loop iteration.

CMS schedule: accumulate saliency EMA every step · `prune_step_blocks` from `prune_start` toward `target_density` · `carve()` once at `compact_step` · ReMoE at `route_start`. Saliency = `score_mode` (taylor / grad / magnitude) EMA, pooled 16×16 tile → 128×128 block (score at tile, execute at block).

ReMoE: routing is *per-token activation-conditioned* on each MORTAR layer (prelude / core / coda). In the looped core, different iterations route differently only because their activations differ — *not* via any iteration-index input.